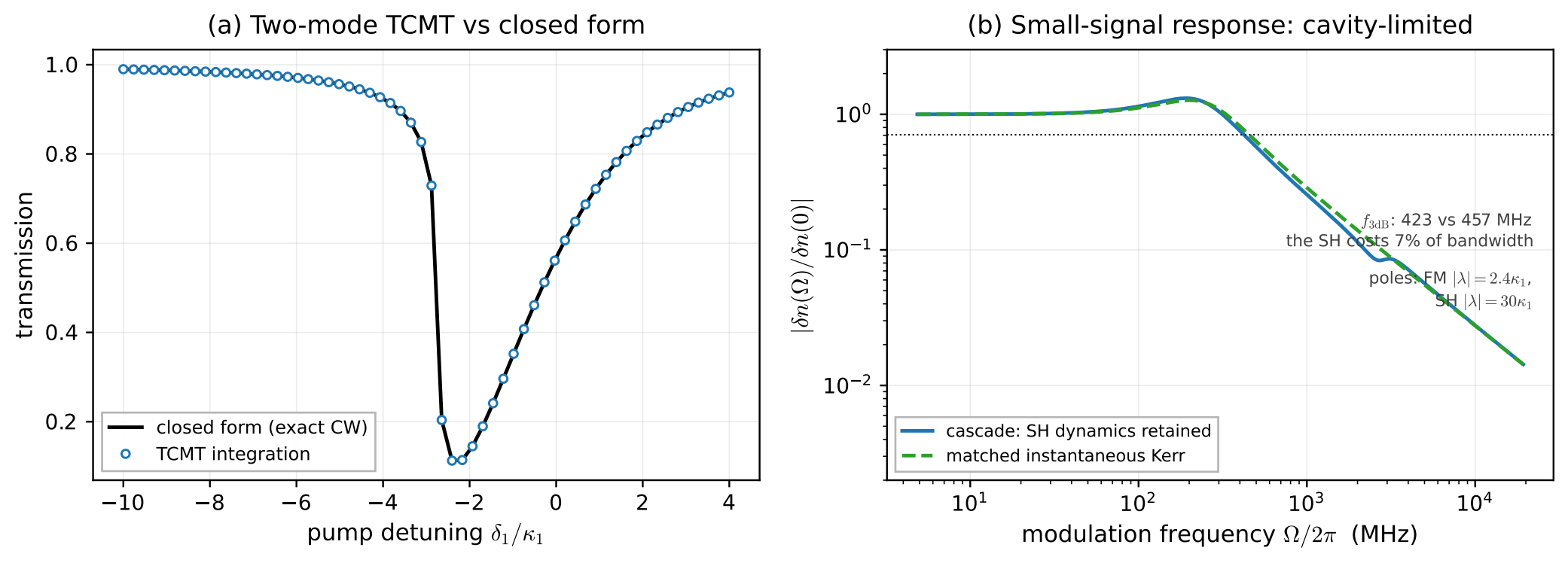
(a) Two-mode TCMT vs closed form
transmission
1.0
0.8
0.6
0.4
0.2
−10
−8
−6
−4
−2
0
2
4
pump detuning δ_1/κ_1
closed form (exact CW)
TCMT integration
(b) Small-signal response: cavity-limited
$|\delta n(\Omega)/\delta n(0)|$
10^0
10^{-1}
10^{-2}
10^1
10^2
10^3
10^4
modulation frequency $\Omega/2\pi$ (MHz)
$f_{3\mathrm{dB}}$: 423 vs 457 MHz
the SH costs 7% of bandwidth
poles: FM $|\lambda| = 2.4\kappa_1$,
SH $|\lambda| = 30\kappa_1$
cascade: SH dynamics retained
matched instantaneous Kerr